A Convergent Discrete Replica Derivative on the Spinor Cone, with the Two-Term Camporesi–Higuchi Spectral Action on $S^3 \times S^1$

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Abstract

We prove one new theorem and record the two continuum results it serves. The theorem concerns a finite-difference discretization of the two-dimensional cone of apex angle $2\pi\alpha$ carrying an anti-periodic (spinor) field: its heat trace $K_n(\alpha, t)$ converges to the continuum heat trace in C^1 in α near $\alpha = 1$, so the replica derivative $dK_n/d\alpha|_{\alpha=1}$, whose mode sum carries an extra weight $(j + \frac{1}{2})$, converges, and that weight costs nothing in the rate (Theorem 3.1). The proof dominates the mode series and its α-derivative by one summable Gaussian sequence. Numerically, the replica combination converges to the continuum value $1/6$, at order $O(a)$ in the radial spacing a on the production grid. The continuum results are these. (i) The Chamseddine–Connes spectral action of the Camporesi–Higuchi Dirac operator on S^3_R and on $S^3_R \times S^1_\beta$ has exactly two power-law terms, up to exponentially small corrections. This is Chamseddine and Connes's result (by Poisson summation); we give an alternative proof through the Bernoulli identity $B_{2k+1}(3/2) = (2k+1)/4^k$, which makes the spectral zeta function vanish at every non-positive integer. The scalar and transverse-traceless tensor heat traces on S^3 do not truncate. (ii) Traced over the rank-4 spinor bundle, the action's Einstein–Hilbert coefficient gives $G = 3\pi/(\phi(1)\Lambda^2)$ for a cutoff with Mellin moments $\phi(s)$. The replica derivative of the black-hole cigar's tip term, with the Dirac cone coefficient $-(1/12)(1/\alpha - \alpha)$ and the fermionic sign of the effective action, gives $S_{\mathrm{BH}} = A\Lambda^2\phi(1)/(12\pi) = A/(4G)$, positive and with the same G. That the two routes agree is the induced-gravity statement of Susskind–Uglum and Frolov–Fursaev, specialised to this spectrum. The absence of higher-curvature terms holds on the conformally flat $S^3 \times S^1$; the cigar is not conformally flat, and there the two routes agree at leading order in Λ. We also show that the second variation of the action on the $(1, 1)$ sector of $SO(4)$ cannot distinguish the diagonal-$SU(2)$ spins $J = 0, 1, 2$, so it cannot by itself isolate a graviton. None of these results uses the GeoVac label lattice: they concern the continuum Camporesi–Higuchi spectrum and a finite-difference cone.

Contents

1 Introduction **2**

2 The spinor cone: continuum and discrete **3**

3 Theorem L6: the replica weight is harmless **4**

4 Spectral action on S^3_R: two-term exactness **6**

5 The product $S^3 \times S^1_\beta$ and its $\beta \to \infty$ limit **8**

6 The truncation is specific to the spinor bundle **8**

7 The Newton constant by two routes **9**
7.1 Action route 9
7.2 Entropy route 10
7.3 Cutoff dependence 11

8 The second variation on the $(1,1)$ sector **12**
8.1 Positive kinetic modes 12
8.2 J-blindness 13

9 Discrete-substrate measurements **14**

10 Open questions **14**

1 Introduction

The spectral action $\mathrm{Tr}\, f(D^2/\Lambda^2)$ of Chamseddine and Connes [6, 9] turns the spectrum of a Dirac operator into a gravitational action. This paper records three results about it on the round three-sphere and on the Euclidean black-hole cigar.

1. **A convergent discrete replica derivative (new).** The Bekenstein–Hawking entropy is obtained by the replica method as a derivative with respect to the apex angle $2\pi\alpha$ of a cone. On a discretized cone, the derivative weights each angular mode by $(j+\frac{1}{2})$, so its convergence does not follow from that of the heat trace. Theorem 3.1 shows that it does converge, in C^1 near $\alpha=1$, and at the heat-trace rate. (INTERNAL THEOREM, with one standard ingredient argued rather than proved, §3; `tests/test_paper51_L6_full.py`.)

2. **Two-term exactness on S^3 and $S^3\times S^1$.** The spectral zeta function of the Camporesi–Higuchi spectrum [5] vanishes at every non-positive integer (Theorem 4.1), so the spectral action has exactly two power-law terms. The result is Chamseddine and Connes's [8]; the Bernoulli route is an alternative proof. (SYMBOLIC PROOF; `tests/test_paper51_zeta_unit_neg_k.py`.)

3. **One Newton constant from two routes.** The Einstein–Hilbert coefficient of the action and the replica derivative of the cigar tip give the same $G=3\pi/(\phi(1)\Lambda^2)$ (§7), once the action is traced over the rank-4 spinor bundle and the entropy is taken with the fermionic sign. (INTERNAL THEOREM, continuum-level; `tests/test_paper51_replica_sbh.py`, `tests/test_paper51_cutoff_mellin.py`.)

Scope. No result in this paper uses the GeoVac label lattice (the finite set of (n,l,m) labels of the S^3 harmonics, [1]). The continuum results use only the Camporesi–Higuchi spectrum $|\lambda_n|=n+3/2$ with multiplicity $2(n+1)(n+2)$, and Theorem 3.1 concerns a finite-difference discretization of a flat cone, which is not the label lattice. The results are offered as spectral geometry of the round three-sphere and the cone, not as consequences of the lattice. Two further limits apply throughout. First, "two terms" and "no higher-curvature corrections" are statements about the conformally flat backgrounds S^3 and $S^3\times S^1$ (§5), not about the spectral action in general. Second, the cutoff function f is external input, as in the Chamseddine–Connes programme; the paper fixes how each prediction depends on it (§7.3) but not its values.

Prior art. The two-term result is Chamseddine and Connes's [8]. The spinor cone coefficient is Kabat's [13] and Fursaev and Miele's [14]. The agreement of the action-route and entropy-route Newton constants for minimally coupled fields is the induced-gravity statement of Susskind

and Uglum [15] and of Frolov and Fursaev [16]; Solodukhin's review records it for scalars and Dirac fermions [17]. What is new here is Theorem 3.1 and the bookkeeping that carries these statements through the Camporesi–Higuchi spectrum and the rank-4 bundle.

Correspondence with the record. The computations were run as a numbered sequence recorded in the repository `CHANGELOG.md`, and other GeoVac papers cite this paper by those labels. They map as follows: G1, G2 and G5 are §4–§5; G3 is §6; G4-1, G4-2, G7, G8 and the cutoff-moment map (G4-5d) are §7–§7.3; G6, its Fierz–Pauli follow-on and the diagnostics GD-1, GD-4 and GD-5 are §8; the discrete-substrate measurements G4-3 and G4-4 and the radial-refinement diagnostic B4 are §2 and §9; Theorem L6 and its rate and target measurements B1 and B2 are §3. The labels F1–F7 name the falsifier panel of the discrete substrate in the test files. The full sprint record, including the threads this version no longer carries, is kept verbatim in `docs/chronicles/paper51_gravity_arc.md`.

The paper is organized as follows. Section 2 sets up the continuum and discrete cones, and Section 3 proves Theorem 3.1. Sections 4–6 treat the spectral action on S^3 and $S^3 \times S^1$. Sections 7–7.3 derive the Newton constant by the two routes and give its cutoff dependence. Section 8 treats the second variation on the $(1,1)$ sector, Section 9 reports the discrete-substrate measurements, and Section 10 lists what is open, including the tests the paper owes.

2 The spinor cone: continuum and discrete

Continuum. Let C_α be the flat two-dimensional cone of apex angle $2\pi\alpha$, cut off at radius R with a Dirichlet condition. Its heat trace has the small-t form

$$K(\alpha, t) = \frac{\alpha\,\pi R^2}{4\pi t}\, c_0 - (\text{perimeter term} \propto \alpha) + c(\alpha) + O(t^{1/2}), \tag{1}$$

with c_0 the number of field components. The area and perimeter terms are linear in α. The apex term $c(\alpha)$ is the Sommerfeld–Cheeger coefficient [12, 11, 10]. For a real scalar it is $+\frac{1}{12}(1/\alpha - \alpha)$. For a two-component Dirac field, with the trace taken over the spinor index, it is the negative of the scalar value [13, 14]:

$$c_{1/2}(\alpha) = -\frac{1}{12}\left(\frac{1}{\alpha} - \alpha\right). \tag{2}$$

(Fursaev and Miele state that the spin-$\frac{1}{2}$ cone coefficient "is just the minus of the same coefficient of the scalar operator", their Eq. (4.13), in agreement with Kabat.) Because the area and perimeter terms are linear in α, the replica combination

$$\mathrm{tip}(t) = \left.\frac{dK}{d\alpha}\right|_{\alpha=1} - K(1, t) \tag{3}$$

keeps only the apex term, and for the spinor it equals $c'_{1/2}(1) = +1/6$ at small t/R^2 (to 2×10^{-8} for $t/R^2 \le 0.05$; at $t/R^2 = 0.1$ the finite-disk value is 0.166520).

Discrete. The discrete cone used here has radial sites $\rho_i = ia$, $i = 1, \ldots, N_\rho$, with $R = N_\rho a$ and a Dirichlet wall, and a Hermitian finite-difference polar Laplacian obtained from the substitution $u = \sqrt{\rho}\, f$. The spinor's azimuthal momenta are $m = (j + \frac{1}{2})/\alpha$ (anti-periodic boundary condition, taken either exactly or by an azimuthal finite difference with N_ϕ sites), and each mode sees the radial operator $L(m) = -\partial_\rho^2 + (m^2 - \frac{1}{4})/\rho^2$. The discrete heat trace on the wedge is denoted $K^{\mathrm{Dirac}}_{\mathrm{wedge}}(\alpha, t)$, and the disk ($\alpha = 1$) heat trace $K^{\mathrm{Dirac}}_{\mathrm{disk}}(t)$.

The discrete tip coefficient. On this substrate the subtraction $\Delta_K(\alpha,t) = K^{\mathrm{Dirac}}_{\mathrm{wedge}}(\alpha,t) - \alpha\, K^{\mathrm{Dirac}}_{\mathrm{disk}}(t)$ recovers Eq. (2) on the deficit side $\alpha < 1$. At $R = 10$, $a = 0.05$, $N_\rho = 200$, the mean recovery over $\alpha \in \{2/5, 1/2, 3/5, 2/3\}$ at $t = 1$ is 99.4%, and the single best cell is $\alpha = 2/5$, $t = 2$, at 1.000021; that cell is where a sweep crosses the target, not a converged value. (MEASURED; `tests/test_paper51_g44_headlines.py`.) A periodic scalar on the same substrate recovers its coefficient $+\frac{1}{12}(1/\alpha - \alpha)$ at only 66.3% on the same four α values and $t = 1$. The scalar has an $m = 0$ mode with an attractive $-1/(4\rho^2)$ term in the u-representation, the spinor has no $m = 0$ mode, and that difference is why the spinor extraction is clean. (An earlier scalar test on a different estimator — the fitted slope of Δ_K against $(1/\alpha - \alpha)$ over $\alpha \in \{1/3, \ldots, 3\}$ with both branches — found 28% of the target at $t = 0.005$, rising to 53% at $t = 1$; the "about one quarter" quoted for it in earlier versions is the $t = 0.005$ value. The three figures are different measurements, not one measurement quoted three ways.)

Excess angle. At $\alpha > 1$ and fixed $a = 0.05$ the recovery plateaus near 67.9%, independent of the azimuthal count. Refining the radial mesh removes it: at $\alpha = 2$, $t = 1$, the recovery rises as $N_\rho = 200 \to 3200$ $(0.675, 0.769, 0.837, 0.885, 0.919)$, with $1 - \text{recovery} \propto \sqrt{a}$ (successive error ratios $1.4085, 1.4143, 1.4164$, near $\sqrt{2}$). The continuum excess-angle coefficient is therefore the plain continuation of Eq. (2), and the fixed-a plateau is a discretization artifact (`tests/test_paper51_owed_guards.py`).

3 Theorem L6: the replica weight is harmless

The replica method needs the derivative of the heat trace with respect to α, and the question for the discrete cone is whether the discrete derivative converges to the continuum one, and at what rate. On the discrete substrate (exact azimuthal momenta $m_{\mathrm{eff}} = (k + \frac{1}{2})/\alpha$), $K(\alpha,t) = 4\sum_{j\geq 0} h((j+\frac{1}{2})/\alpha, t)$ with $h(m,t)$ the radial heat trace, and $\frac{dK}{d\alpha}|_1 = -4\sum_j (j+\frac{1}{2})h'(j+\frac{1}{2})$. The per-mode *replica contribution* $C_j = -4(j+\frac{1}{2})h'(j+\frac{1}{2}) > 0$ carries the $(j+\frac{1}{2})$ *replica weight*, heavier than the bare heat trace.

Write $n = (N_\phi, N_\rho, a)$ for the discrete truncation (N_ϕ azimuthal modes, N_ρ radial sites, spacing a, $R = N_\rho a$ fixed), and $g_j^{(n)}(\alpha) := 4\, h_n\big((j+\frac{1}{2})/\alpha,\, t\big)$, so that $K_n(\alpha,t) = \sum_{j\geq 0} g_j^{(n)}(\alpha)$ and the per-mode replica contribution is $C_j(\alpha) = -g_j^{(n)\prime}(\alpha)$. The continuum objects g_j, K are the $n \to \infty$ limits.

Theorem 3.1 (L6: replica-weight-harmless convergence — INTERNAL THEOREM, Lemma 3.3(b) argued; `tests/test_paper51_L6_full.py`). *Fix $t > 0$ and $\delta \in (0,1)$. As $n \to \infty$ the maps $\alpha \mapsto K_n(\alpha,t)$ converge to $\alpha \mapsto K(\alpha,t)$ in $C^1([1-\delta, 1+\delta])$. In particular*

$$\lim_{n\to\infty} \frac{dK_n}{d\alpha}\bigg|_{\alpha=1} = \frac{dK}{d\alpha}\bigg|_{\alpha=1} = \frac{d}{d\alpha}\bigg|_{\alpha=1} \lim_{n\to\infty} K_n, \tag{4}$$

so $\mathrm{tip}_n(t) \to \mathrm{tip}(t) = 1/6$. *The convergence rate of the replica derivative is the per-mode rate of the radial heat trace and its m-derivative: the replica weight $(j+\frac{1}{2})$ adds only a super-polynomially small tail and does not degrade it.*

The proof uses Lemmas 3.3 and 3.4; the second is the new content. Lemma 3.2 concerns geometric faithfulness of the truncation, is **[OPEN]**, and is not used.

Lemma 3.2 (Propinquity backbone; **[OPEN]**, not used). *For each fixed α near 1, the truncated constant-warp metric spectral triples $\mathcal{T}_{n,\alpha}$ converge to the continuum tip triple $\mathcal{T}_\alpha$ in the van Suijlekom state-space Gromov–Hausdorff distance, $d_{\mathrm{GH}}(\mathcal{T}_{n,\alpha}, \mathcal{T}_\alpha) \leq C_3\, \gamma_n \to 0$, with C_3 truncation-independent.*

The azimuthal circle and the horizon sphere have known reconstruction maps (the Toeplitz circle of Connes and van Suijlekom, [3]; Berezin–Toeplitz quantization of S^2), but the disk-with-cone factor does not: its $1/\rho^2$ term couples the radial and azimuthal parts, it is a manifold with boundary, and it is not a group. Paper 53 [4] proves state-space convergence on the flat plane by a positive squared-window reconstruction and translation invariance, and leaves the disk and the cone open; the cone $\alpha \neq 1$ has no translation group. The compressed disk operator system has propagation number 2 at the truncations tested (MEASURED). The lemma would certify faithfulness at each α; it says nothing about the α-derivative, which is what Theorem 3.1 needs.

Lemma 3.3 (Pointwise heat-trace convergence)**.** *(a) For each fixed (α, t), $K_n(\alpha, t) \to K(\alpha, t)$ as $n \to \infty$, with error bounded by a radial discretisation term $O(a^p)$ and super-polynomially small azimuthal and radial truncation terms. The radial operator $L(m) = -\partial_\rho^2 + (m^2 - \frac{1}{4})/\rho^2$ on $(0, R]$ has, for every $m = (j + \frac{1}{2})/\alpha > 0$, eigenfunctions $\sim \rho^{|m|+1/2}$ vanishing at the apex, so $\rho = 0$ needs no boundary condition and the only boundary datum is the Dirichlet wall at $\rho = R$. (b) For each fixed j, $g_j^{(n)}$ and $g_j^{(n)\prime}$ converge to g_j and g_j' uniformly on $[1 - \delta, 1 + \delta]$.*

Status. Part (a) is finite-difference convergence for a regular Sturm–Liouville problem: the spinor has no $m = 0$ mode, so the apex is centrifugally screened. Part (b) follows from the convergence of the discrete eigenvalues and eigenvectors of $L(m)$, locally uniformly in m, together with the Hellmann–Feynman formula $\partial_m \lambda_k = 2m\langle\rho^{-2}\rangle_k$. We argue it here; we do not prove it in detail. The exponent is measured (B1 below): $p = 1$ on the production grid $\rho_i = ia$, and $p = 2$ on a boundary-corrected grid. (Part (a) is the bounded-interval norm-resolvent statement recorded in the archived Paper 47.)

Lemma 3.4 (The replica weight is harmless — new)**.** *There is a constant $c(\delta) = t/((1+\delta)^2 R^2) > 0$ and, for each (t, δ), a summable Gaussian-dominating sequence*

$$G_j \;=\; C(t,\delta)\,(j+\tfrac{1}{2})^2\, e^{-c(\delta)(j+\frac{1}{2})^2}, \qquad \sum_{j\geq 0} G_j < \infty, \tag{5}$$

such that $\left|g_j^{(n)}(\alpha)\right| \leq G_j$ and $\left|g_j^{(n)\prime}(\alpha)\right| \leq G_j$ for all $\alpha \in [1 - \delta, 1 + \delta]$ and all n large. Hence $\sum_j g_j^{(n)}$ and $\sum_j g_j^{(n)\prime}$ converge uniformly on $[1 - \delta, 1 + \delta]$.

Proof. The centrifugal floor obeys $\lambda_0(m) \geq m^2/R^2$ (continuum Bessel-zero bound $j_{m,1} \geq m$; on the lattice the floor is at least as steep, $\lambda_0(m) \geq c_a\, m^2/R^2$ with $c_a \to 1^+$ as $a \to 0$, confirmed numerically below at $c_a \approx 1.16$). The excited gaps $\lambda_k(m) - \lambda_0(m)$ grow like $k^2\pi^2/R^2$ uniformly in m (the high modes see the box, not the centrifugal barrier), so the relative partition function $Z(t) = \sum_k e^{-t(\lambda_k - \lambda_0)}$ is bounded uniformly in m. Therefore

$$h(m,t) \leq Z(t)\, e^{-tm^2/R^2}, \qquad |h'(m,t)| \leq t\, C'\, (m/R^2)\, Z(t)\, e^{-tm^2/R^2}, \tag{6}$$

the second from $|\partial_m \lambda_k| = 2m\langle\rho^{-2}\rangle_k \leq C'm/R^2$ (Hellmann–Feynman; the screened modes concentrate near $\rho \sim R$ so $\langle\rho^{-2}\rangle = O(R^{-2})$). For $\alpha \in [1 - \delta, 1 + \delta]$ and $m_j = (j + \frac{1}{2})/\alpha$ one has $m_j \geq (j + \frac{1}{2})/(1 + \delta)$, hence $e^{-tm_j^2/R^2} \leq e^{-c(\delta)(j+\frac{1}{2})^2}$, while $|g_j^{(n)\prime}| = 4|h'(m_j)|(j + \frac{1}{2})/\alpha^2$ carries a polynomial factor $\leq (j + \frac{1}{2})^2/(1 - \delta)^2$. Both $|g_j^{(n)}|$ and $|g_j^{(n)\prime}|$ are thus bounded by (5), which is summable because a Gaussian beats any polynomial. The Weierstrass M-test gives uniform convergence of both series. □

Proof of Theorem 3.1. By Lemma 3.3(b) each term $g_j^{(n)\prime}$ converges to g_j' uniformly on $[1-\delta, 1+\delta]$, and by Lemma 3.4 every term is bounded by G_j, uniformly in n and α, with $\sum_j G_j < \infty$. Dominated convergence for series (Tannery's theorem) then gives $\sum_j g_j^{(n)\prime} \to \sum_j g_j'$ uniformly in α, and likewise for $\sum_j g_j^{(n)}$. Since each $K_n(\cdot, t)$ is C^1, the uniform limit of derivatives is the

derivative of the limit: $K(\cdot,t) \in C^1([1-\delta, 1+\delta])$, $K_n \to K$ in C^1, and evaluating at $\alpha = 1$ exchanges $\lim_n$ with $d/d\alpha$. For the rate, split the series at J: the terms $j < J$ converge at the per-mode rate, and the tail $\sum_{j \geq J} G_j$ is super-polynomially small in J, so the weight $(j+\frac{1}{2})$ enters only through a tail that is negligible at any polynomial rate. The tip $\mathrm{tip}_n = dK_n/d\alpha|_1 - K_n(1,t)$ inherits the convergence. □

The dominating exponent $c(\delta) \to t/R^2$ as $\delta \to 0$ is the envelope slope measured below.

Numerical verification. Unless stated, $R = 10$ and $t = 1$; a lean tridiagonal solver is cross-checked bit-exact against the production class `DiscreteDiskDiracSpectral` of `geovac/gravity/warped_dirac.py`. *(i) Gaussian envelope.* $C_j \sim \exp(\mathrm{slope} \cdot (j+\frac{1}{2})^2)$, measured slope -0.01129 vs predicted $-t/R^2 = -0.01000$; the replica weight is crushed, C_j/C_0 falling from 4.7 at j=4 to 1.6×10^{-20} at j=64, and the azimuthal partial sum reaches the total to machine precision by $K_{\mathrm{az}} \approx 48$ (tail/total $= 4 \times 10^{-13}$). *(ii) Uniformity.* The envelope slope across $\alpha \in [0.9, 1.1]$ lies in $[-0.01134, -0.01123]$, so the domination is uniform in α. The ratio $(K_{\mathrm{fine}} - K_{\mathrm{coarse}})/K_{\mathrm{coarse}}$ at $(a, a') = (0.05, 0.025)$ and $t = 10$ varies by 1.3% (coefficient of variation; 0.23% at $t = 1$) across $\alpha \in \{0.90, 0.95, 1.00, 1.05, 1.10\}$. *(iii) Derivative consistency.* At a single truncation the per-mode analytic $dK_n/d\alpha = 41.501652$ and a finite difference of K_n, 41.501681, agree to 2.9×10^{-5}. This checks the derivative implementation at fixed n; it does not test the interchange of $\lim_n$ with $d/d\alpha$, which is what the theorem proves.

The rate and the target (B1, B2). *(B1, rate.)* The discrete lowest eigenvalue $\lambda_0(m; a)$ against the exact continuum value $(j_{m,1}/R)^2$ (Bessel zero) converges at order $q = 1.00$ at every $m \in \{\frac{1}{2}, \frac{3}{2}, \frac{5}{2}\}$, and the tip itself at order $p = 0.97$, so $p = 1$ on the production grid. The cause is the placement of the Dirichlet wall, not the centrifugal term: the grid $\rho_i = ia$ puts the wall at $(N_\rho + 1)a = R + a$, displaced by $O(a)$. Two checks confirm it: the case $m = \frac{1}{2}$, where the centrifugal term vanishes, still converges at $O(a)$; and the grid $a = R/(N_\rho + 1)$, with the wall exactly at R, restores $O(a^2)$ (error ratio 4). *(B2, target.)* Linear extrapolation in a of the tip gives 0.166654 (two points) and 0.166582 (least squares), against $1/6 = 0.166667$, that is -0.008% and -0.05%, with the α-derivative taken exactly (Hellmann–Feynman). On a separate panel at $t = 10$ ($a \in \{0.1, 0.05, 0.025, 0.0125, 0.00625\}$) the tip's error against 1/6 is -5.32, -2.73, -1.41, -0.75, -0.42%, converging from below to the finite-disk continuum value at $t/R^2 = 0.1$, 0.166520 (-0.088% from 1/6); a central difference in α with step 0.1 biases the tip by $+0.5\%$ and would show a spurious sign change (`tests/test_paper51_owed_guards.py`).

4 Spectral action on S^3_R: two-term exactness

The Camporesi–Higuchi Dirac operator on the unit S^3 has spectrum

$$|\lambda_n^{CH}| = n + 3/2, \qquad g_n = 2(n+1)(n+2), \qquad n \geq 0, \tag{7}$$

where g_n counts both signs of the eigenvalue [5]. The spectral action of a spectral triple is

$$S[D] = \mathrm{Tr}\, f(D^2/\Lambda^2) = \sum_n g_n\, f(\lambda_n^2/\Lambda^2), \tag{8}$$

where f is a smooth positive cutoff function and Λ is a mass scale. On S^3_R (radius R), $\lambda_n = (n + 3/2)/R$, so $\lambda_n^2/\Lambda^2 = (n+3/2)^2/(R\Lambda)^2$, and the action depends only on the dimensionless combination $u = R\Lambda$. We work with the Gaussian cutoff $f(x) = e^{-x}$ until Section 7.3, where we generalize to arbitrary f. For the Gaussian cutoff, $\phi(s) := \int_0^\infty f(x) x^{s-1} dx = \Gamma(s)$, so $\phi(k) = (k-1)!$ for integer k and $\phi(1/2) = \sqrt{\pi}$, $\phi(3/2) = \sqrt{\pi}/2$.

Theorem 4.1 (Spectral zeta identity at non-positive integers — SYMBOLIC PROOF; `tests/test_paper51_zeta_unit_neg_k.py`). *For the Camporesi–Higuchi spectrum on unit S^3, the spectral zeta function*

$$\zeta_{\text{unit}}(s) := \sum_{n\geq 0} g_n (n+3/2)^{-2s} \tag{9}$$

satisfies $\zeta_{\text{unit}}(-k) = 0$ for every integer $k \geq 0$.

Proof. $\zeta_{\text{unit}}(s) = 2\sum_{n\geq 0}(n+1)(n+2)(n+3/2)^{-2s}$, which equals $2\sum_{n\geq 0}[(n+3/2)^2 - 1/4](n+3/2)^{-2s}$. This separates as $\zeta_{\text{unit}}(s) = 2\zeta_H(2s-2, 3/2) - \frac{1}{2}\zeta_H(2s, 3/2)$, where ζ_H is the Hurwitz zeta. By standard Hurwitz functional equations, $\zeta_H(-m, a) = -B_{m+1}(a)/(m+1)$ for non-negative integer m. At $s = -k$:

$$\zeta_{\text{unit}}(-k) = -\frac{2B_{2k+3}(3/2)}{2k+3} + \frac{B_{2k+1}(3/2)}{2(2k+1)}.$$

From the shift formula $B_n(x+1) = B_n(x) + nx^{n-1}$ and the identity $B_{2k+1}(1/2) = 0$ (standard for all $k \geq 0$), one obtains $B_{2k+1}(3/2) = (2k+1)/4^k$. Substituting: the first term gives $-2(2k+3)/[4^{k+1}(2k+3)] = -1/(2\cdot 4^k)$ and the second gives $(2k+1)/[4^k \cdot 2(2k+1)] = 1/(2\cdot 4^k)$. Their sum is zero at every $k \geq 0$. □ □

The identity in Theorem 4.1 forces the Mellin-transform expansion of $\operatorname{Tr} f(D^2/\Lambda^2)$ on S^3_R to truncate at low order.

Corollary 4.2 (Two-term exactness of the spectral action — SYMBOLIC PROOF; `tests/test_paper51_zeta_unit_neg_k.py`). *For Gaussian cutoff on S^3_R, the spectral action equals*

$$S(R, \Lambda) = \phi(3/2)(\Lambda R)^3 - \frac{1}{4}\phi(1/2)(\Lambda R) + O(e^{-\Lambda^2 R^2}). \tag{10}$$

On the round S^3_R there are no further power-law terms at any order.

The corresponding extremum equation is $\frac{d}{du}[\phi(3/2)u^3 - \frac{1}{4}\phi(1/2)u] = 0$, giving $3\phi(3/2)u^2 = \frac{1}{4}\phi(1/2)$, or $u^2_{\text{crit}} = \phi(1/2)/(12\phi(3/2))$. For Gaussian cutoff this gives $u^2_{\text{crit}} = (\sqrt{\pi})/(12\cdot\sqrt{\pi}/2) = 1/6$, so $R_{\text{crit}}\Lambda = 1/\sqrt{6}$. This extremum belongs to the two-term form; the exact mode sum $\operatorname{Tr} f(D^2/\Lambda^2)$ is monotonic in R.

Remark 4.3 (Relation to Chamseddine–Connes). Chamseddine and Connes [8] showed, by Poisson summation over the Camporesi–Higuchi spectrum, that the spectral action on S^3 and on $S^3 \times S^1$ is given by two terms up to rapidly decaying corrections, so that all higher Seeley–DeWitt coefficients vanish, and checked a_4 and a_6 by a local heat-kernel computation. The Bernoulli argument of Theorem 4.1 is an alternative algebraic proof of the same all-order vanishing: the identity $B_{2k+1}(3/2) = (2k+1)/4^k$ forces exact pairwise cancellation in the Hurwitz decomposition at every non-positive integer. Under the standard volume normalization the two Dirac coefficients are $a_0^{D^2} = 4\pi^2$ and $a_1^{D^2} = -2\pi^2$; with the scalar series $a_k^{\Delta} = 2\pi^2/k!$ of Section 6, all lie in $\bigoplus_k \pi^{2k}\cdot\mathbb{Q}$, consistent with the period statements of Fathizadeh and Marcolli for Robertson–Walker spectral actions [7] (Paper 32 [2], § VIII).

Remark 4.4 (Odd spheres). The identity $\zeta_{\text{unit}}(-k) = 0$ also holds for the Camporesi–Higuchi spectra of S^5 and S^7, checked symbolically through $k = 7$ (`tests/test_paper51_owed_guards.py`). The mechanism is the same: the degeneracy polynomials (degree 4 and 6) are even in the shifted eigenvalue $n + d/2$, and the Bernoulli multi-step formula $B_{2m+1}(p+1/2) = (2m+1)\sum_{l=0}^{p-1}(l+1/2)^{2m}$ produces the cancellation. On S^{2m+1} the spectral action then has exactly $m+1$ power-law terms, $\Lambda^{2m+1}, \Lambda^{2m-1}, \ldots, \Lambda$; on S^3 these are the volume and scalar-curvature terms. On a round sphere every local curvature invariant is a power of the constant scalar curvature, so the term count does not by itself say which invariants (curvature squared, Weyl squared, Gauss–Bonnet) a general background would carry; Section 5 states what can be said.

5 The product $S^3 \times S^1_\beta$ and its $\beta \to \infty$ limit

The Euclidean thermal product geometry $S^3 \times S^1_\beta$ has Dirac operator $D = D_{S^3} \otimes \sigma_1 + I \otimes \sigma_2 \, (-i\partial_\tau)$ with $\tau \in [0, \beta)$, acting on the rank-4 bundle $\mathbb{C}^2 \otimes \mathbb{C}^2$ (two-component S^3 spinors tensored with $\mathbb{C}^2$; odd-dimensional S^3 has no chirality operator, so a rank-2 product is not a Dirac operator). Then $D^2 = D^2_{S^3} \otimes I - \partial^2_\tau$ and the heat kernel factorizes with a factor 2:

$$K_{S^3 \times S^1}(t) = 2\, K_{S^3}(t) \cdot K_{S^1}(t) = 2\, K_{S^3}(t) \cdot \frac{\beta}{\sqrt{4\pi t}}. \tag{11}$$

Corollary 5.1 (4D spectral action with exact two-term structure — INTERNAL THEOREM)**.** *The Chamseddine–Connes spectral action on $S^3_R \times S^1_\beta$ for Gaussian cutoff equals*

$$S(R, \beta, \Lambda) = \frac{\beta R^3}{2}\Lambda^4 - \frac{\beta R}{4}\Lambda^2 + O(\textit{exp small}). \tag{12}$$

What "no higher-curvature terms" means here. The Weyl tensor of $S^3 \times S^1$ vanishes, as does the Gauss–Bonnet density there [8]; the background is conformally flat. In general the fourth Seeley–DeWitt coefficient of the spectral action contains the square of the Weyl curvature and a topological term [8]. So Corollary 5.1 says that on this background the action is the Einstein–Hilbert term plus a cosmological term, with nothing else. It does not say that the spectral action is pure Einstein gravity on other backgrounds. In particular the Euclidean black-hole cigar of Section 7 is not conformally flat, and there the Weyl-squared term is present (§7, Wald paragraph).

The $\beta \to \infty$ limit. Sending $S^1_\beta \to \mathbb{R}_\tau$ replaces K_{S^1} by the density $K_{\mathbb{R}}(t) = 1/(2\sqrt{\pi t})$, and the action density per unit Euclidean time is

$$s(R, \Lambda) = \frac{R^3}{2}\Lambda^4 - \frac{R}{4}\Lambda^2, \tag{13}$$

on $S^3 \times \mathbb{R}$, the Euclidean Einstein static universe. Its two-term form has the same formal extremum $R_{\text{crit}}\Lambda = 1/\sqrt{6}$ as on S^3, where it takes the value

$$s_{\min} = -\frac{\Lambda}{6\sqrt{6}}. \tag{14}$$

(SYMBOLIC; `tests/test_paper51_dS_vacuum.py`, which computes the density from the spectrum on the rank-4 bundle.) This is an observation about the two-term density, not a vacuum-selection result: the exact mode sum is monotonic in R.

6 The truncation is specific to the spinor bundle

The two-term exactness of Sections 4 and 5 is specific to the Camporesi–Higuchi spinor bundle on S^3. We exhibit the heat traces of three sectors: the spinor sector (Dirac), the scalar Laplacian, and transverse-traceless (TT) symmetric tensors.

Dirac sector (spinor bundle).

$$K_{D^2}(t) = \frac{\sqrt{\pi}}{2}t^{-3/2} - \frac{\sqrt{\pi}}{4}t^{-1/2} + O(e^{-\pi^2/t}), \tag{15}$$

where the exponentially-small correction comes from Jacobi ϑ_2 modular transformation on the half-integer-shifted Camporesi–Higuchi spectrum.

Scalar sector. The bosonic Laplacian on unit S^3 has spectrum $\lambda_n^\Delta = n(n+2)$ with multiplicity $(n+1)^2$.

Theorem 6.1 (Closed-form Seeley–DeWitt coefficients for scalar Laplacian on unit S^3 — INTERNAL THEOREM (classical θ_3/Poisson asymptotics; algebra symbolic, closed form matched at 50 dps); `tests/test_paper51_scalar_ak.py`). *The heat trace satisfies*

$$K_\Delta(t) = \frac{\sqrt{\pi}}{4}\,\frac{e^t}{t^{3/2}} + O(e^{-\pi^2/t}) = \sum_{k=0}^{\infty} a_k^\Delta\, t^{k-3/2}\cdot(4\pi)^{-3/2}, \tag{16}$$

with $a_k^\Delta = 2\pi^2/k!$ *for all* $k \geq 0$.

The proof uses the Jacobi ϑ_3 modular transformation on the integer-spectrum bosonic sum, exploiting the structural difference between half-integer (Dirac) and integer (Laplacian) spectra.

TT-tensor sector. The rough Laplacian $\nabla^*\nabla$ on transverse-traceless symmetric 2-tensors on unit S^3 has spectrum $(n+1)(n+3)-2$, $n \geq 1$ (the Lichnerowicz operator is $\nabla^*\nabla + 6$ on this sector). Writing the spectrum as m^2-3 ($m \geq 3$) with multiplicity $2(m^2-4)$, the heat trace is

$$K_{TT}(t) = e^{3t}\left[\frac{\sqrt{\pi}}{2}t^{-3/2} - 4\sqrt{\pi}t^{-1/2} + 4\right] + 6e^{2t} + O(e^{-\pi^2/t}), \tag{17}$$

which is pure integer ϑ_3 content: the full-lattice sum is a ϑ_3 series, and the $+4$ and $6e^{2t}$ terms are the excluded $m = 1, 2$ modes. (`tests/test_paper51_tt_heat_trace.py`.)

Remark 6.2 (Structural distinction). The two-term exactness holds only for the Dirac sector (half-integer spectrum, ϑ_2 content). The scalar Laplacian and the TT sector (integer spectra, ϑ_3 content) have infinite Seeley–DeWitt series, $a_k^\Delta = 2\pi^2/k!$ for the scalar. The two-term Einstein–Hilbert plus cosmological form of Sections 4 and 5 therefore belongs to the spinor sector; a scalar or tensor field on the same background carries the full tower.

7 The Newton constant by two routes

7.1 Action route

Matching the spectral action (12) to Einstein–Hilbert plus a cosmological constant,

$$S_{\rm EH} = -\frac{1}{16\pi G_{\rm eff}}\int R_{\rm scalar}\sqrt{g} + \frac{\Lambda_{cc}}{8\pi G_{\rm eff}}\int\sqrt{g}, \tag{18}$$

with the volume forms on $S^3_R \times S^1_\beta$ giving $\mathrm{Vol} = 2\pi^2 R^3\beta$ and $\int R_{\rm scalar}\sqrt{g} = 12\pi^2 R\beta$, the coefficient matching gives

$$\Lambda^2 \text{ term}: \quad G_{\rm eff} = 3\pi/\Lambda^2, \tag{19}$$

$$\Lambda^4 \text{ term}: \quad \Lambda_{cc} = 6\Lambda^2. \tag{20}$$

Proposition 7.1 (Newton constant and cosmological constant — INTERNAL THEOREM). *The spectral action on* $S^3_R\times S^1_\beta$, *traced over the rank-4 bundle with Gaussian cutoff, produces Newton constant* $G_{\rm eff} = 3\pi/\Lambda^2$ *(Planck-scale) and cosmological constant* $\Lambda_{cc} = 6\Lambda^2$. *(`tests/test_paper51_cutoff_mellin.py`, which derives both from the spectral sum.)*

Setting $\Lambda = M_{\rm Planck}$ gives $\Lambda_{cc}/M^2_{\rm Planck} = 6$, against an observed value of order 10^{-120}. This is the standard cosmological-constant problem of the Chamseddine–Connes programme [9]; the present construction neither adds to it nor resolves it.

7.2 Entropy route

Sign conventions. We use one set throughout, that of Solodukhin's review [17]. The Euclidean effective action is $W(\alpha) = -\ln Z(\alpha)$ on the cone of apex angle $2\pi\alpha$, and the replica entropy is

$$S = \Big(\alpha\,\frac{\partial}{\partial\alpha} - 1\Big)W(\alpha)\Big|_{\alpha=1}. \tag{21}$$

For a Dirac field $Z = \det D$, so $W = -\frac{1}{2}\ln\det D^2 = +\frac{1}{2}\int (ds/s)\,\mathrm{Tr}\,e^{-sD^2}$ (regulated): the heat trace enters with a positive weight, the opposite of the bosonic sign [13, 17]. The spectral action $\mathrm{Tr}\,f(D^2/\Lambda^2) = \int_0^\infty g(s)\,K(s/\Lambda^2)\,ds$, with f the Laplace transform of $g \geq 0$, enters the same way, and we take $W = I_E := \mathrm{Tr}\,f(D^2/\Lambda^2)$. With this sign the negative $\Lambda^2 R$ term of Eq. (12) gives $G > 0$, as above; Solodukhin notes that the negative sign of the Dirac curvature coefficient and the fermionic sign of the effective action together give a positive contribution to $1/G$.

The cigar. The Euclidean Schwarzschild cigar is

$$ds^2 = d\rho^2 + \rho^2 d\phi^2 + r(\rho)^2 d\Omega_2^2, \tag{22}$$

with ϕ-period 2π for a smooth tip and $2\pi\alpha$ off-shell, and horizon sphere of radius $r(0) = r_h$. Near the tip, at constant warp $r = r_h$, the squared Dirac operator is $D^2_{C_\alpha} \otimes I + I \otimes D^2_{S^2}/r_h^2$ on a rank-4 bundle (two-component cone spinors tensored with two-component S^2 spinors), and the heat trace factorizes. The S^2 Dirac spectrum is $|\lambda_n| = (n+1)/r_h$ with multiplicity $4(n+1)$, $n \geq 0$, so

$$K_{S^2}(t) = \sum_{n\geq 0} 4(n+1)e^{-(n+1)^2 t/r_h^2} = \frac{2r_h^2}{t} - \frac{1}{3} + O(t). \tag{23}$$

This is an infinite Seeley–DeWitt series (integer spectrum, ϑ_3 content), with first curvature coefficient $a_1^{S^2} = -4\pi/3$ in the $(4\pi t)^{-1}$ normalization. The cone tip contributes $c_{1/2}(\alpha)\,K_{S^2}(t)$ to the heat trace, Eq. (2).

The entropy. Integrating the tip term against the cutoff (Proposition 7.3 below) gives

$$I_E^{\mathrm{tip}}(\alpha) = c_{1/2}(\alpha)\,\big[2r_h^2\Lambda^2\,\phi(1) - \tfrac{1}{3}f(0)\big] = -\frac{r_h^2\Lambda^2\,\phi(1)}{6}\left(\frac{1}{\alpha} - \alpha\right) + O(\Lambda^0). \tag{24}$$

The tip term vanishes at $\alpha = 1$ and the bulk terms are linear in α, so only $\partial_\alpha I_E^{\mathrm{tip}}$ survives in Eq. (21), and with $c'_{1/2}(1) = +1/6$,

$$S_{\mathrm{BH}} = \frac{r_h^2\Lambda^2\,\phi(1)}{3} - \frac{f(0)}{18} = \frac{A\Lambda^2\,\phi(1)}{12\pi} + O(\Lambda^0), \tag{25}$$

with horizon area $A = 4\pi r_h^2$. The leading term is positive. Matching $S_{\mathrm{BH}} = A/(4G)$ gives

$$G = \frac{3\pi}{\phi(1)\,\Lambda^2}, \tag{26}$$

the action-route value of Proposition 7.1 and Proposition 7.2. (`tests/test_paper51_replica_sbh.py` computes both routes from their spectra at the Gaussian cutoff and checks that they agree.) The two inputs to the agreement are the rank-4 trace of Section 5 and the total-trace cone coefficient $\frac{1}{12}|1/\alpha - \alpha|$ for the two-component spinor; a trace over a rank-2 bundle, or a per-component reading of the cone coefficient, would each introduce a factor of 2.

Why the routes agree, and where. For a minimally coupled field, the one-loop entropy and the one-loop renormalization of $1/G$ come from the same heat-kernel coefficient: this is Susskind and Uglum's observation [15], and it underlies the induced-gravity account of black-hole entropy [16]. Solodukhin's review states that each minimally coupled field, bosonic or fermionic, contributes positively and proportionally to both [17]. The agreement above is that statement for the Camporesi–Higuchi spectrum. Wald's Noether-charge entropy [18] gives the same conclusion for a pure Einstein–Hilbert Lagrangian, but the cigar is not conformally flat, and there the spectral action carries a Weyl-squared term at order Λ^0 (§5). That term changes the Wald entropy at order Λ^0, so the identification of the two G's holds at the leading order Λ^2, not to all orders. Nonminimally coupled fields (gauge fields, gravitons) break the proportionality [13, 17]; the statement here is for the Dirac field only.

7.3 Cutoff dependence

For an arbitrary cutoff f, define the Mellin moments

$$\phi(s) := \int_0^\infty f(x)\, x^{s-1}\, dx. \tag{27}$$

Proposition 7.2 (Cutoff-dependent gravity predictions; `tests/test_paper51_cutoff_mellin.py`). *For arbitrary smooth positive cutoff f with finite Mellin moments $\phi(1), \phi(2)$, the spectral action on $S^3_R \times S^1_\beta$ produces*

$$G_{\text{eff}}(f,\Lambda) = \frac{3\pi}{\phi(1)\,\Lambda^2}, \tag{28}$$

$$\Lambda_{cc}(f,\Lambda) = \frac{6\,\phi(2)}{\phi(1)}\,\Lambda^2, \tag{29}$$

$$R_{\text{crit}}\,\Lambda = \sqrt{\frac{\phi(1)}{6\,\phi(2)}}. \tag{30}$$

Table 1: Gravity predictions for three cutoff functions.

Cutoff $f(x)$	$\phi(1)$	$\phi(2)$	$G_{\text{eff}}\Lambda^2$	Λ_{cc}/Λ^2	$R_{\text{crit}}\Lambda$
Gaussian e^{-x}	1	1	3π	6	$1/\sqrt{6}$
Polynomial e^{-x^2}	$\sqrt{\pi}/2$	$1/2$	$6\sqrt{\pi}$	$6/\sqrt{\pi}$	$\sqrt{\sqrt{\pi}/6} \approx 0.544$
Sharp $\Theta(1-x)$	1	$1/2$	3π	3	$1/\sqrt{3}$

No combination of these is independent of the cutoff: $G_{\text{eff}}\Lambda^2 = 3\pi/\phi(1)$, $\Lambda_{cc}/\Lambda^2 = 6\,\phi(2)/\phi(1)$, $G_{\text{eff}}\Lambda_{cc} = 18\pi\,\phi(2)/\phi(1)^2$, and $R_{\text{crit}}\Lambda = \sqrt{\phi(1)/(6\,\phi(2))}$. The construction fixes which moment each quantity carries; the cutoff function, and hence the values, are external input, as in the Chamseddine–Connes programme [6, 9]. In these terms the cosmological-constant problem is the requirement $\phi(2)/\phi(1)^2 \approx 10^{-124}$ with $\phi(1) = O(1)$, which no natural cutoff meets.

Proposition 7.3 (Sector-wise Mellin moment map; `tests/test_paper51_cutoff_mellin.py`). *Let f be a cutoff function with regulated Mellin moments $\phi(s) = \int_0^\infty f(x)\, x^{s-1}\, dx$, and let the Chamseddine–Connes spectral action on the Euclidean Schwarzschild cigar be decomposed into bulk and topological-tip sectors via the replica method. Then:*

(i) *The cosmological-constant sector (proportional to $a_0 \sim t^{-d/2}$) scales with $\phi(d/2) = \phi(2)$ in $d = 4$.*

(ii) *The Einstein–Hilbert sector (proportional to $a_2 \sim t^{-(d-2)/2}$) scales with $\phi((d-2)/2) = \phi(1)$ in $d = 4$.*

(iii) *The topological tip sector (the Sommerfeld–Cheeger contribution). On the bare 2D cone the tip term is t-independent, so in the spectral action it contributes $\frac{1}{6}f(0)$, independent of the Mellin moments. On the 4D cigar it multiplies the Dirac transverse kernel $K_{S^2}(t) = 2r_h^2/t - 1/3 + O(t)$, so its leading contribution to S_{BH} is $(r_h^2\Lambda^2/3)\,\phi(1)$, the Einstein–Hilbert moment, and the subleading piece is $-f(0)/18$.*

Proof. Write the cutoff as a Laplace transform, $f(x) = \int_0^\infty e^{-sx} g(s)\, ds$, so that $S = \mathrm{Tr}\, f(D^2/\Lambda^2) = \int_0^\infty g(s)\, K(s/\Lambda^2)\, ds$ with $K(t) = \mathrm{Tr}\, e^{-tD^2}$. A heat-trace term $c\, t^{-\beta}$ with $\beta > 0$ contributes $c\,\Lambda^{2\beta}\int_0^\infty g(s) s^{-\beta} ds = c\,\Lambda^{2\beta}\phi(\beta)/\Gamma(\beta)$, and a t^0 term contributes $c\, f(0)$. The Seeley–DeWitt expansion $K(t) \sim \sum_n a_n\, t^{(n-d)/2}$ then gives (i) and (ii) for the $n = 0$ and $n = 2$ bulk coefficients.

For the replica tip sector: the Sommerfeld–Cheeger contribution to the heat trace on a cone of apex angle $2\pi\alpha$ is $\Delta K^{\mathrm{SC}}(\alpha, t) = c(\alpha) \cdot K_{S^{d-2}}(t)$, where $c(\alpha) = -(1/12)(1/\alpha - \alpha)$ for spinors (Eq. (2)) and $K_{S^{d-2}}(t)$ is the transverse-sphere heat kernel; its replica derivative is the constant $1/6$. On the bare 2D cone the tip term is t^0 and contributes $f(0)/6$. On the 4D cigar, $\frac{1}{6}(2r_h^2/t - 1/3)$ contributes $\frac{1}{3}r_h^2\Lambda^2\phi(1) - f(0)/18$; the leading term is the $A\Lambda^2\phi(1)/(12\pi)$ of Eq. (25). □ □

Remark 7.4 (The proper-time estimator orders the cutoffs differently)**.** The discrete measurements of Section 9 weighted the substrate's tip with the proper-time functional $\int (dt/t)\, f(t\Lambda^2)\, \mathrm{tip}(t)$, which for a constant tip is a log-regulated $\phi(0)$ and orders the cutoffs sharp > polynomial > Gaussian (the ordering is pinned in `tests/test_paper51_cutoff_mellin.py`). That is a property of the estimator, not of the spectral action, in which the bare 2D tip carries $f(0)$ and the 4D leading tip carries $\phi(1)$. A naive expectation that the entropy would carry $\phi(2)$ is rejected by the same measurement.

8 The second variation on the $(1, 1)$ sector

This section asks what the spectral action alone says about spin-two perturbations of the round S^3 Dirac operator. The answer is a necessary condition and a limit.

8.1 Positive kinetic modes

The Camporesi–Higuchi eigenspaces carry the $SO(4) = SU(2)_L \times SU(2)_R$ irreps $((n+1)/2, n/2)$ and $(n/2, (n+1)/2)$, and the products of an eigenspace with itself contain the irrep $(j_L, j_R) = (1, 1)$ (dimension 9) for $n \geq 1$. For a perturbation $\delta D = X$ of the Dirac operator D_0 supported within one eigenspace of eigenvalue λ (a *within-sector* mode), the Gaussian spectral action's second variation is

$$A_\lambda = a_\lambda \left(\frac{4\lambda^2}{\Lambda^4} - \frac{2}{\Lambda^2} \right), \qquad a_\lambda = e^{-\lambda^2/\Lambda^2}. \tag{31}$$

Modes connecting different eigenspaces (*cross-sector* modes) include the inner-automorphism tangents $V = i[X, D_0]$, whose entries $(\lambda_b - \lambda_a)X_{ab}$ vanish within a sector; so the inner gauge acts only across sectors.

Proposition 8.1 (Positive within-sector modes — MEASURED, $n_{\max} \leq 3$)**.** *$A_\lambda > 0$ exactly when $\lambda^2 > \Lambda^2/2$. At $\Lambda^2 = 6$ the within-sector $(1, 1)$ modes at $|\lambda| = 5/2, 7/2, 9/2$ have $A_\lambda = +0.127, +0.133, +0.066$, and a five-point finite-difference stencil reproduces the formula. For $\Lambda^2 > 2\lambda_{\min}^2 = 12.5$ the lowest block is negative. Positivity is a necessary condition for spin-two dynamics, and it holds only on the infrared side $\Lambda^2 < 2\lambda_{\min}^2$. (`tests/test_paper51_j_blindness.py` checks the formula against the stencil.)*

Degree-of-freedom count. In the continuum the nine components of $(1,1)$ reduce to two transverse-traceless polarizations once diffeomorphism gauge (four) and constraints (three) are removed. At finite truncation this reduction does not occur inside a within-sector block, because the inner gauge acts only across sectors (zero within-sector gauge entries, checked at $n_{\max} = 1,2,3$). The reduction can only appear in a limit in which the cross-sector directions assemble into continuum diffeomorphisms; no such limit is established here.

8.2 J-blindness

Under the diagonal $SU(2) \subset SU(2)_L \times SU(2)_R$,

$$(1,1) = J{=}0 \;\oplus\; J{=}1 \;\oplus\; J{=}2 \tag{32}$$

with dimensions $1, 3, 5$. If $(1,1)$ is read as the symmetric traceless four-tensors, these are the h_{00} component, the h_{0i} components and the traceless spatial h_{ij}; that reading requires a dictionary between δD and $\delta g_{\mu\nu}$, which this paper does not supply.

Proposition 8.2 (J-blindness of $S^{(2)}$ — `tests/test_paper51_j_blindness.py`)**.** *The Gaussian spectral-action second variation $S^{(2)}$ restricted to the $(1,1)$ subspace of* $\mathrm{Herm}(H_{n_{\max}})$ *has identical eigenvalue spectra across $J = 0, 1, 2$. For cross-sector modes, eigenvalue multiplicities scale as* $\dim(J) = 1{:}3{:}5$*; for within-sector modes, after Hermitianization, as* $1{:}1{:}3$*. The same argument applies to every irrep in a block, so the spectral action assigns the helicity-two irreps $(2,0) \oplus (0,2)$ the same eigenvalues as $(1,1)$.*

Proof. The spectral action $S = \mathrm{Tr}(f(D^2/\Lambda^2))$ is invariant under unitary conjugation $D \mapsto UDU^*$, hence under the $SO(4)$ symmetry of D_0 and in particular under the diagonal $SU(2)$. The second variation $S^{(2)}$ is therefore equivariant under diagonal $SU(2)$. By Schur's lemma, $S^{(2)}$ is proportional to the identity on each irreducible J-component within a given sector pair (a,b), with proportionality constant

$$w(\lambda_a, \lambda_b) = \begin{cases} e^{-\lambda_a^2/\Lambda^2}(4\lambda_a^2/\Lambda^4 - 2/\Lambda^2) & \lambda_a = \lambda_b, \\ \dfrac{(\lambda_a + \lambda_b)^2}{\Lambda^4}\,\dfrac{\Lambda^2(a_a - a_b)}{\lambda_b^2 - \lambda_a^2} - \dfrac{a_a + a_b}{\Lambda^2} & \lambda_a \neq \lambda_b, \end{cases} \tag{33}$$

where $a_k = e^{-\lambda_k^2/\Lambda^2}$. This weight depends only on the eigenvalues λ_a, λ_b of D_0, not on the angular quantum numbers or on J. □

The test checks the eigenvalue-set and multiplicity identities at $n_{\max} = 1, 2$, and the weight (33) against a five-point stencil at the 10^{-4} level at $n_{\max} = 1$ (stencil noise sets that floor).

Which irrep carries the graviton. The $(1,1)$ irrep has $|j_L - j_R| = 0$: it is a helicity-zero sector. The transverse-traceless tensor harmonics on S^3 lie in $(j{+}2, j) \oplus (j, j{+}2)$, lowest $(2,0) \oplus (0,2)$, with $|j_L - j_R| = 2$ (their multiplicity $2(m^2 - 4)$ is that of Section 6). The within-sector blocks do contain $(2,0) \oplus (0,2)$ (checked at $n = 1, \ldots, 4$), and by the Proposition they carry the same A_λ as $(1,1)$, so the positivity of Proposition 8.1 extends to them under the same condition on Λ. What the Proposition rules out is any separation: the spectral action alone cannot distinguish the transverse-traceless modes from the trace or longitudinal ones, nor any J from another. That separation needs the identification of δD with a metric perturbation (the Chamseddine–Connes dictionary), which breaks $SO(4)$ to the $SO(3)$ that distinguishes them; whether such an identification exists on a truncation is open. (The local polarization space is $(1,1)$ while the global transverse-traceless harmonics are $|\Delta j| = 2$, and the dictionary may move irrep assignments; that is why the question is open rather than settled negatively.)

9 Discrete-substrate measurements

The continuum statements of Sections 4–7.3 were re-measured on a discretized cigar. These are sprint-scale measurements, and the end-to-end discrete entropy is not converged.

Substrate. The near-horizon cigar is discretized as the cone of Section 2 tensored with the S^2 Dirac spectrum truncated at $l \le l_{\max}$, with warp $r(\rho) = r_h$ (constant) or $r_h\sqrt{1+(\rho/r_h)^2}$ (smooth tip). The continuum Dirac operator on the warped cigar is

$$D_{\text{cigar}} = D_{D^2} \otimes I_{S^2} + \gamma^5_{D^2} \otimes \frac{D_{S^2}}{r(\rho)} + \frac{r'(\rho)}{r(\rho)}\,\gamma^\rho \otimes I_{S^2}. \tag{34}$$

Checks that pass. (All MEASURED; `tests/test_paper51_L6_full.py` F1–F7 and `tests/test_paper51_g44_headlines.py`.)

- At constant warp the heat trace factorizes, $K_{\text{cigar}} = K_{D^2} \cdot K_{S^2}$, to machine precision; the variable-warp operator reduces to the constant-warp one at constant warp to $\sim 10^{-16}$; the chirality grading anticommutes with the disk Dirac operator.
- The spinor-to-scalar heat-trace ratio is 1.9998 (rank two).
- At $t = 0.1$, $N_\phi = 192$, $N_\rho = 200$, $a = 0.05$, the leading coefficient is $a_0^{\text{Dirac}} = 1.992$. Against the bulk value 2.0 this omits the Dirichlet perimeter term; against the continuum value of the same estimator with the perimeter term (1.888 at $R = 10$, $t = 0.1$) the substrate is 5.5% high, and its $a \to 0$ sweep has not reached it. The value 1.992 is a crossing of the bulk value, not a recovery.
- As $r_h \to 0$ the variable-warp heat trace tends to the singular-tip cone ($r(\rho) = \rho$) heat trace, to 6×10^{-8} relative at $r_h = 10^{-3}$, quadratically in r_h; this is by construction, since the lattice excludes $\rho = 0$. It is not the naive ceiling $4(l_{\max}+1)(l_{\max}+2)\,K_{\text{disk}}$, because the lattice spacing fixes a finite S^2 mass at the apex: the substrate has two scales, a and r_h.
- At smooth-tip warp the factorization fails by a positive amount that grows as r_h shrinks.

The tip coefficient and the replica derivative. The spinor tip coefficient and its excess-angle behaviour are those of Section 2; the replica derivative is Theorem 3.1 with the measurements B1 and B2.

What is not converged. Integrating the discrete replica derivative against a cutoff to obtain S_{BH} on the substrate gives values that depend strongly on Λ and on the t-grid; at one cell ($r_h = 2$, $\Lambda = 2$) an 8-point grid gives 0.85 of the continuum $r_h^2\Lambda^2/3$ and a 13-point grid gives 1.96. The cause is that the substrate's tip profile peaks at a t fixed by a and r_h, not by Λ; the cure is radial refinement, not a different cutoff. The end-to-end discrete S_{BH} is therefore not established, and this paper claims only the continuum Eq. (25) and the convergence of the replica derivative at fixed t.

10 Open questions

Q1: the discrete entropy. Theorem 3.1 controls the replica derivative at each fixed t. An end-to-end discrete S_{BH} needs the convergence uniform over the t-window of the cutoff integral, the joint variable-warp and conical Dirac operator in the continuum limit, and a converged extraction (Section 9). The propinquity backbone, Lemma 3.2, is open.

Q2: the graviton dictionary. Proposition 8.2 shows the spectral action cannot by itself separate the spin-two modes. Whether a truncation supports an identification of δD with $\delta g_{\mu\nu}$, and whether the $9 \to 2$ reduction appears in a limit, are open.

Q3: the cutoff. The cutoff function is external input. No mechanism examined here selects one, and none produces the $\phi(2)/\phi(1)^2 \approx 10^{-124}$ that the observed cosmological constant would need.

Tests. B1, B2, B4, the continuum tip and a_0, the odd-sphere identity, the rank-4 trace, the total-trace cone coefficient, the coefficient of variation of the discretization error across α (1.3% at $t = 10$, 0.23% at $t = 1$) and propagation number 2 of the compressed disk operator system are backed by `tests/test_paper51_owed_guards.py`.

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